Solution (i) Equation of line AB, $A(0,5)$ and $B(4,1)$

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y-5}{1-5} = \frac{x-0}{4-0}$$

$$4(y-5) = -4x \Rightarrow y-5 = -x$$

$$x + y - 5 = 0$$

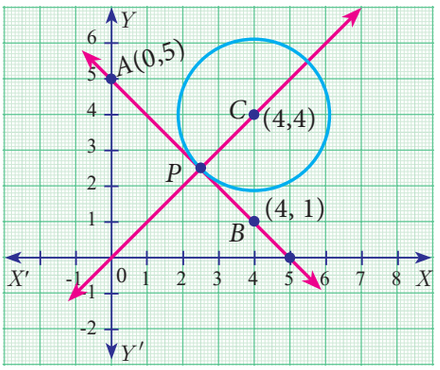


Fig. 5.40

(ii) The equation of a line which is perpendicular to the line $AB: x+y-5=0$ is $x-y+k=0$

Since it is passing through the point $(4,4)$, we have

$$4 - 4 + k = 0 \Rightarrow k = 0$$

The equation of a line which is perpendicular to AB and through C is $x-y=0$

(iii) The coordinate of the point of contact P of the tangent line AB with the circle is point of intersection of lines.

$x+y-5=0$ and $x-y=0$

solving, we get $x=\frac{5}{2}$ and $y=\frac{5}{2}$

Therefore, the coordinate of the point of contact is $P\left(\frac{5}{2},\frac{5}{2}\right)$.

Thinking Corner

1. Find the number of point of intersection of two straight lines.
2. Find the number of straight lines perpendicular to the line $2x-3y+6=0$.

Activity 6

Find the equation of a straight line for the given diagrams

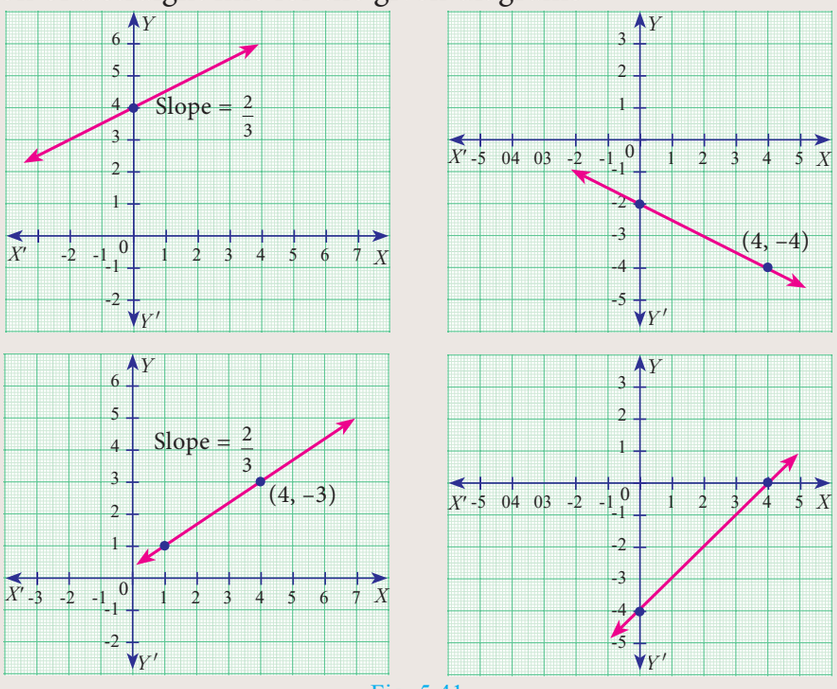


Fig. 5.41

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